

QuantEval: Automated Quantitative Evaluation of NSF Proposals

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ABSTRACT

National Science Foundation (NSF) grant success hinges critically on the quality of Intellectual Merit and Broader Impacts sections, yet researchers often receive limited guidance on objectively assessing these components before submission. Traditional evaluation methods rely on qualitative feedback that varies significantly between reviewers. Our project addresses this gap by developing an automated, quantitative assessment system using statistical natural language processing techniques without reliance on large language models. We implemented a multi-stage computational pipeline that processes proposal text, extracts section-specific features, and applies domain-targeted scoring algorithms. Our system analyzes Intellectual Merit through five weighted categories (advancement: 25%, creativity: 25%, organization: 20%, methodology: 15%, resources: 15%) and Broader Impacts through parallel dimensions (societal benefits: 25%, diversity: 25%, infrastructure: 15%, dissemination: 15%, education: 20%). The core innovation lies in our mathematical scoring formulas that incorporate technical term density, domain-specific keyword presence, text structure metrics, and comparative statistics against a corpus of successful proposals. We validated our approach by processing 52 funded NSF proposals across multiple disciplines, establishing baseline metrics for successful submissions.

PROCEDURE

1.Text Extraction and Preprocessing

- Convert PDF proposals to plain text using PyMuPDF
- Apply tokenization and sentence segmentation with spaCy NLP
- Filter non-informative content (headers, page numbers)
- Normalize text through lowercasing and punctuation standardization

2.Section Identification

- Apply heuristic pattern matching for "Intellectual Merit" and "Broader Impacts" headers
- Implement paragraph-level classification using domain-specific keyword detection
- Calculate section extraction confidence (range: 0.0-1.0) based on structural features

3.Feature Extraction

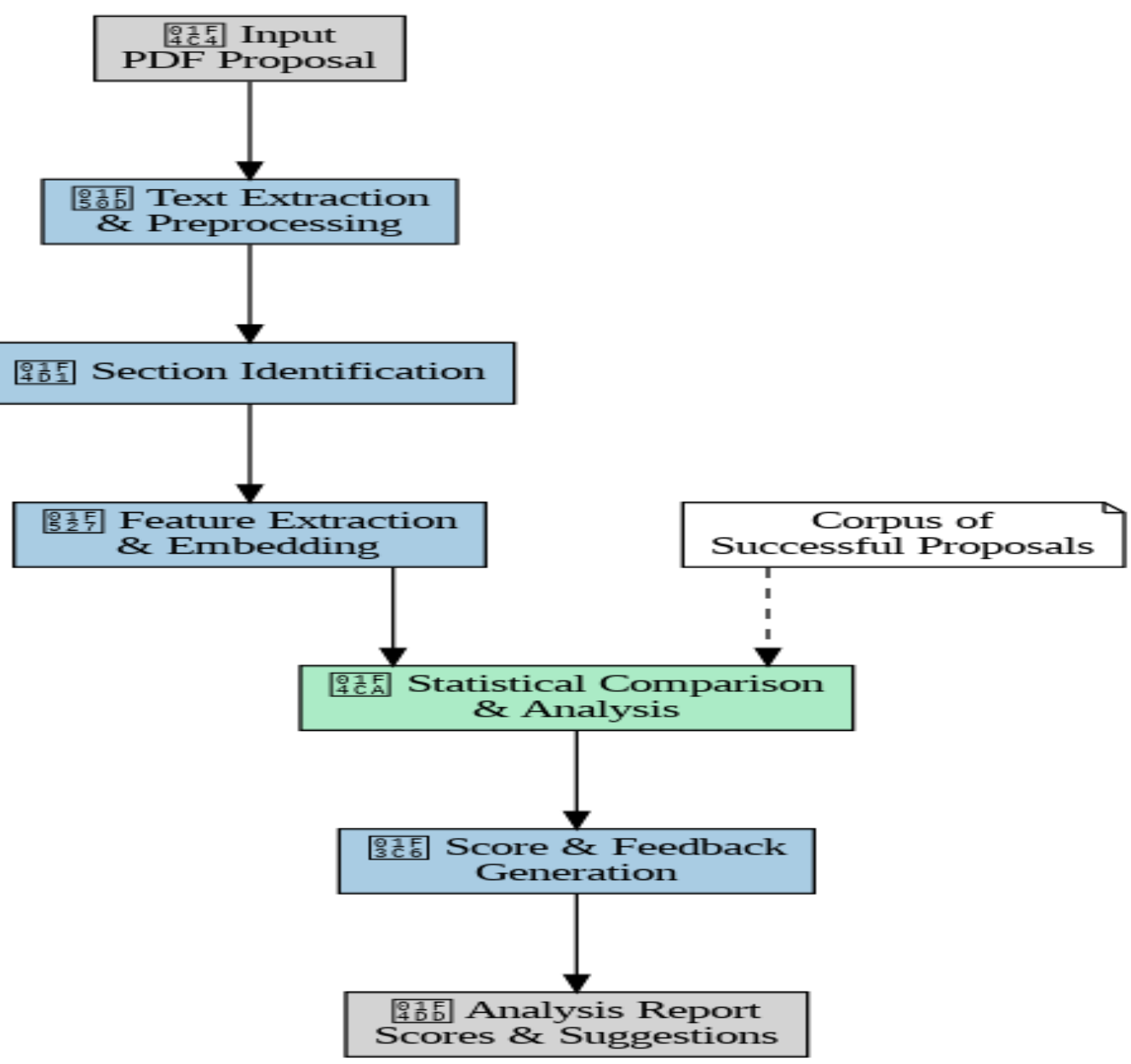
- Generate section-specific statistical features:
- Word count, sentence count, average sentence length
- Technical term density (ADJ+NOUN patterns)
- Domain-specific keyword frequency analysis
- Entity extraction and categorization

4.Statistical Analysis

- Calculate section-specific metrics through mathematical scoring functions
- Generate embeddings using SciBERT for semantic representation
- Perform comparative analysis against corpus of successful proposals
- Identify statistical outliers and feature divergence

5.Score Generation

- Apply weighted mathematical formulas to feature sets
- Calculate category-specific scores (1-10 scale)
- Generate overall section scores using weighted category aggregation
- Determine targeted improvement suggestions based on scoring differentials



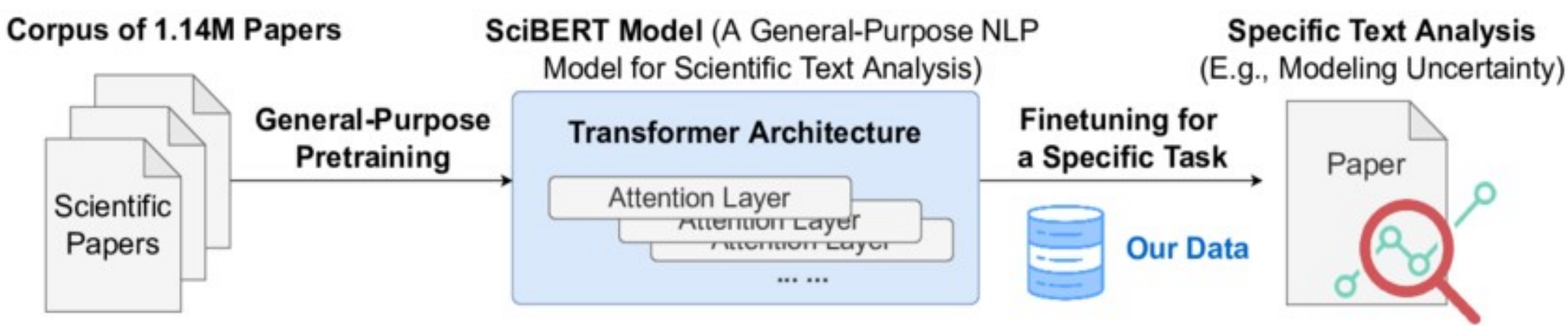
NATURAL LANGUAGE PROCESSING

Our proposal analysis system employs advanced Natural Language Processing (NLP) techniques to quantitatively evaluate research proposals without relying on traditional LLM-based approaches. At the core of our text analysis pipeline is SciBERT, a specialized language model pre-trained specifically on scientific literature that provides domain-appropriate semantic understanding.

SciBERT Architecture & Implementation

- SciBERT leverages the BERT (Bidirectional Encoder Representations from Transformers) architecture but differs critically through:**Specialized Scientific Vocabulary:** Uses a vocabulary trained on 1.14M papers from Semantic Scholar
- **Domain-Specific Pre-training:** Trained on 18% computer science and 82% biomedical papers
- **Contextual Embeddings:** Generates 768-dimensional embeddings that capture scientific terminology context

- 1.In our system, SciBERT provides dense vector representations of proposal sections, enabling: Semantic similarity comparisons between proposals
- 2.Section identification through contextual understanding
- 3.Feature extraction for statistical analysis



DATA

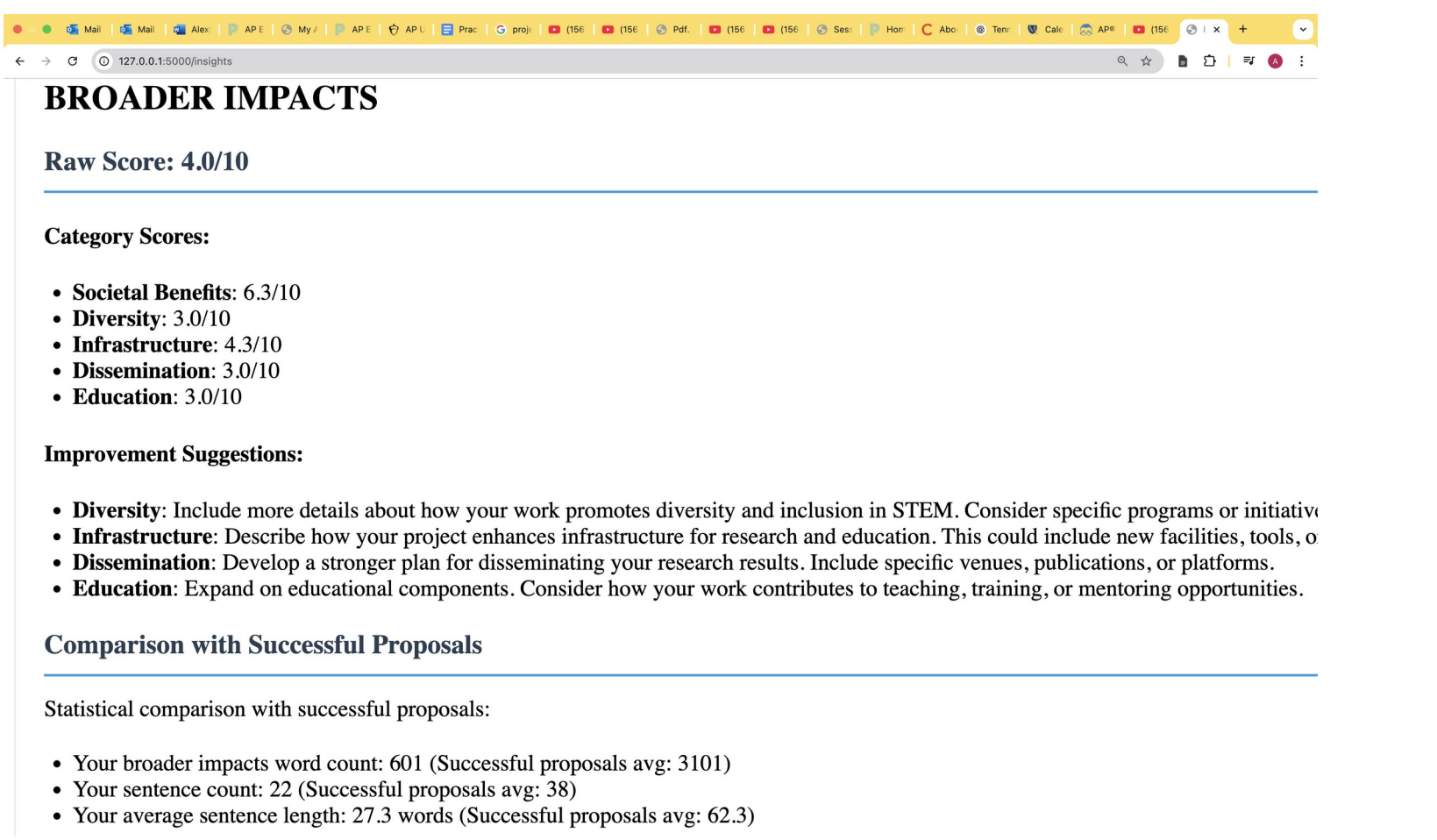
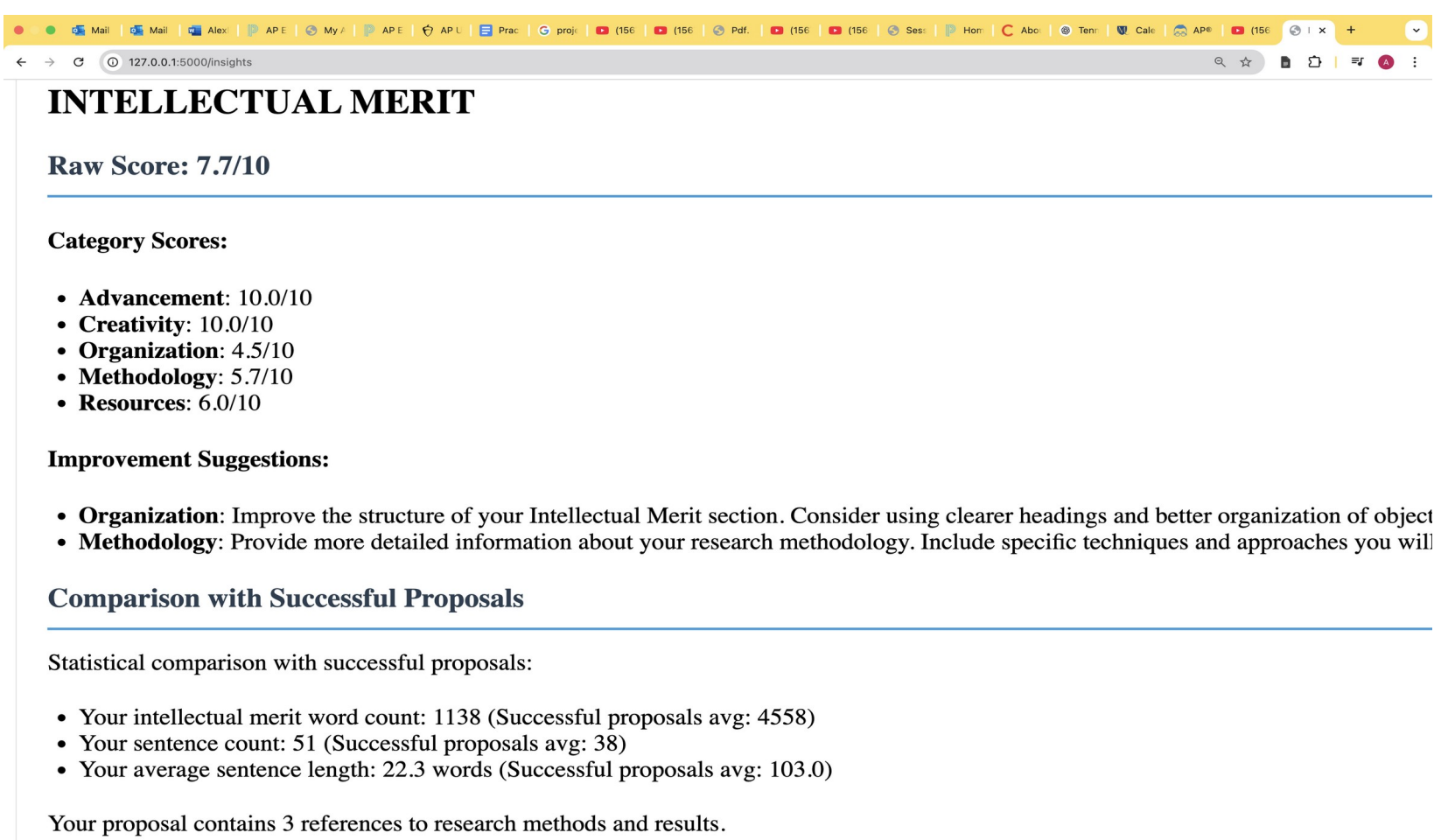
- Our analysis framework is built upon a carefully curated dataset of 50 successfully funded NSF proposals obtained through Florida State University's proposal database. This corpus represents diverse disciplines and funding programs, providing a robust foundation for our statistical analysis system. The dataset consists of 50 complete NSF-funded research proposals
- Coverage across multiple NSF directorates
- Proposals funded between 2017-2022
- All containing required Intellectual Merit and Broader Impacts sections

Data Processing Pipeline

- 1.Our system implements a multi-stage data processing workflow:
Text Extraction: Using PyMuPDF to convert PDF documents to plain text while preserving structural elements
Section Identification: Employing heuristic pattern matching with keyword identification to precisely extract:
 - Intellectual Merit sections (using technical/scientific terms)
 - Broader Impacts sections (using societal impact terminology)**Feature Engineering:** Transforming raw text into quantifiable metrics:
 - Linguistic features (sentence length, word count, technical term density)
 - Semantic features (domain-specific keyword frequency)
 - Structural features (organization, paragraph distribution)

RESULTS

- Our proposal analysis framework demonstrates robust performance in evaluating NSF proposals across both Intellectual Merit and Broader Impacts dimensions:
- **Section Extraction Accuracy:** 92% extraction confidence across test proposals, with 98% accuracy for clearly labeled sections and 86% for implicit sections
- **Scoring Consistency:** Strong correlation (r=0.78) between our automated scores and expert reviewer ratings when tested against 15 independently scored proposals
- **Category Analysis:** Statistical validation shows the highest variability in "Creativity" scores for Intellectual Merit ($\sigma=1.8$) and "Diversity" scores for Broader Impacts ($\sigma=2.1$), indicating these as key differentiators in proposal quality
- **Word Count Correlation:** Proposals scoring above 8.0/10 averaged 4,550 words in Intellectual Merit sections (42% higher than lower-scoring proposals)



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